

Choosing a Subsystem

I am in Software Engineering. So, I chose the following subsystem with my partner for the Software category.

Facility Security Software: A centralized software platform that provides dashboards for security camera access and entry-point status monitoring, enforces security protocols, manages sensor and camera control, and handles alert and breach responses using camera-based monitoring supported by computer vision.

The other pair of our team are in Electrical Engineering and they chose the following.

Surveillance & Security Systems: Physical security infrastructure including cameras, biometric scanners, motion detectors, and AI-based monitoring hardware.

These two subsystems are closely related, as the Surveillance & Security Systems provide real-time data and signals that are processed, controlled, and visualized by the Facility Security Software. This subsystem pairing was reviewed and approved by the TA.

Subsystem Problem Definition

1. What is the problem your subsystem addresses in the CSA facility?

Answer: The Facility Security Software addresses the following issues within the CSA facility:

- Unauthorized access to live and recorded camera footage
- Unauthorized entry into restricted rooms or laboratories
- Incorrect sensor data or improper camera control behaviour
- Failure of alerts or alarms to trigger during actual security breaches

2. What are the objectives for this subsystem?

Answer: The objectives of the Facility Security Software subsystem are:

- To prevent unauthorized access to live and recorded security camera footage
- To enforce strict access control for staff and visitors based on assigned permissions and facility zones

- To ensure sensor and camera data accuracy through redundancy and validation mechanisms
- To detect security breaches promptly and trigger alerts or alarms immediately upon detection

3. What are 3–5 constraints your subsystem design must incorporate?

Answer:

- The number of personnel allowed to/have access to the video footage recording is less than 10 in a week.
- The total number of sensors (cameras, motion, access) that can be in place for the entire facility must be less than 1000.
- The total power drawn by the facility due to security camera, sensor or alert/breach response must be less than 100 kW.
- The alert system must pass 99.9% time during testing.

4. Who are the key stakeholders, and how are they impacted (positively or negatively)?

Answer:

- **Facility Security Guards:** Require fast, reliable access to live feeds and recordings, along with limited staff and visitor identification information for monitoring purposes
- **Facility Staff:** Must have seamless access to authorized rooms while being restricted from unauthorized areas
- **Facility Visitors:** Must be prevented from entering restricted areas unless accompanied by authorized staff

Individual Quantified Design

For the Facility Security Software design, I prioritize minimizing upfront capital costs by reducing the initial number of installed cameras. This decision is made at the expense of higher operational costs, such as employing additional security guards to manually inspect areas not covered by cameras.

Design Parameters

- **Permanent data storage:** ~250–300 GB/day
- **CPU usage:** ~120–150 CPU-hours/day at ~2.4–3.0 GHz
- **Electrical power consumption by software subsystem alone:** ~25–30 kW
- **Video Retention Period for Critical Cameras:** 60-90 days of retained footage for high-security zones
- **Cloud latency:** 1–2 seconds for alerts

Justifications for design parameters

Permanent data storage (~250–300 GB/day): This accounts for continuous recording from a limited set of high-security cameras at moderate resolution, along with metadata, access logs, and computer-vision event records required for audits and investigations.

CPU usage (~120–150 CPU-hours/day at ~2.4–3.0 GHz): This level supports access control enforcement, event aggregation, selective computer-vision processing, and alert generation without overprovisioning compute resources.

Electrical power consumption by software subsystem alone (~25–30 kW): This reflects the power required for servers, storage, networking equipment, and redundancy overhead, remaining well within the facility's overall security power budget.

Video retention period for critical cameras (60-90 days): Retaining footage only from high-security zones balances federal compliance against storage cost and data minimization requirements.

How Long Can You Keep CCTV Footage?

You may only store footage as **long** as it's necessary for its intended purpose.

- 30 days is standard for residential or small retail
- **60**–90 days for high-risk **commercial** areas
- Indefinitely **ONLY** if an incident occurred and it's needed for **legal** or investigative purposes

You must document and communicate this **retention policy** to all stakeholders.

Figure 1: Federal law regarding security camera footage retention period found from <https://northwestsecurity.ca/canada/ontario/canadian-cctv-privacy-laws-official-compliance-guidance/>

Cloud latency (1–2 seconds for alerts): This ensures near-real-time breach detection and response while allowing limited cloud processing or logging without compromising operational security.

Conceptual System Description

The Facility Security Software is designed as a centralized system that continuously collects and processes data from cameras and sensors across the facility. This system operates 24/7 to monitor access, validate sensor data, detect unauthorized activity, and trigger alerts or alarms whenever a security breach is identified.

Program-specific course to get reference from:

This subsystem design is informed by the Bell–LaPadula security model, studied in SE 3RA3 (Software Requirements and Security Considerations). The model enforces confidentiality through ‘No Read Up and No Write Down’ rules, ensuring that users with lower security clearance cannot access higher-level camera footage, and users with higher clearance cannot improperly transfer sensitive data to lower security levels.

Gantt Chart

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETED
Subsystem Identification and High-Level Design (DS Wk 2)	1	2	1	2	100%
Cost Estimation and Financial Model Development (DS Wk 3)	8	2	0	0	0%
Project Management and Cost-Benefit Analysis (DS Wk 4)	15	2	0	0	0%
Applying Engineering Economics – Individual Co-op Case Study 1 (DS Wk 5)	22	7	0	0	0%
Economic Evaluation of Design Alternatives (DS Wk 6)	29	2	0	0	0%
Applying Engineering Economics – Economic Refinement and Robustness Analysis (DS Wk 7)	43	2	0	0	0%
Applying Engineering Economics – Individual Depreciation Case Study 2 (DS Wk 8)	50	7	0	0	0%
Applying Engineering Economics – Risk Management and Uncertainty Identification (DS Wk 9)	57	2	0	0	0%
Sensitivity Analysis (DS Wk 10)	64	2	0	0	0%
Final Report Submission (DS Wk 11)	71	14	0	0	0%
Final Team Interview (DS Wk 12)	78	7	0	0	0%
Main Project Self-Reflection (DS Wk 12)	78	7	0	0	0%

Period Highlight: 1

Plan Duration

Actual Start

% Complete

Actual (beyond plan)

% Complete (beyond plan)

